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Drug resistance in *Candida albicans* isolates and related changes in the structural domain of Mdr1 protein

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ABSTRACT

Background: The increasing azole drug resistance in fungal pathogens poses a pressing threat to global health care. The coexistence of drug-resistant *Candida albicans* with tuberculosis patients and the failure of several drugs to treat *C. albicans* infection extend hospital stay, economic burden, and death. The misuse or abuse of azole-derived antifungals, chronic use of TB drugs, different immune-suppressive drugs, and diseases like HIV, COVID-19, etc., have aggravated the situation. So it is vital to understand the molecular changes in drug-resistant genes to modify the treatment to design an alternative mechanism.
Method: *C. albicans* isolated from chronic tuberculosis patients were screened for antifungal sensitivity studies using disk diffusion assay. The multidrug-resistant *C. albicans* were further screened for molecular-level changes in drug resistance using MDR1 gene sequencing and compared with Gen bank data of similar species using the BLAST tool.
Results: The investigation proved that the isolated *C. albicans* from TB patients are significantly resistant to the action of six drugs. The molecular changes in MDR1 genes showed differences in seven nucleotide base pairs that interfered with the efflux pump.

Introduction

The advent of antimicrobials is a gift for human beings to encounter pathogenic microbes to live longer and live healthier. In recent years the pathogenic microbes have evolved molecular adaptations to withstand the inhibiting effects of antimicrobials. Antimicrobial resistance happens when germs like bacteria and fungi develop the ability to defeat the drugs designed to kill them. That means the germs are not killed and continue to grow. So antibiotic-resistant germs have to be checked to prevent them from becoming superbugs. The drug's failure to treat antibiotic-resistant infections extends hospital stays and creates an additional economic burden and toxic alternatives. Antimicrobial resistance [AMR] has become a global challenge for people at any stage of life, animal husbandry, and agriculture practices. Nearly 2.8 million people are affected by antibiotic-resistant bacteria or fungi in the USA, and more than 35,000 people die every year [1]. According to WHO, antimicrobial resistance is an emerging threat to global health and development. It requires worldwide cooperation to achieve sustainable development goals. According to WHO, AMR is one of the top 10 global public health deteriorators. The abuse and overuse of antimicrobials are the prime factors behind developing drug-resistant pathogens [2]. Among the different drug-resistant microbes, fungal species are not an exception. Antifungal resistance is an increasing problem with the fungus *Candida*, a yeast that belonged to Ascomycetes. The genus *Candida* includes more than 150 species. *Candida* species is a common etiological agent of fungal infections in humans. *Candida* species have diverse macroscopic and microscopic characteristics. *Candida* infections may resist antifungal drugs, particularly the azole class of drugs. The fluconazole and echinocandin drugs resistant *Candida* infections have cornered treatment options [3]. After COVID-19 Fungal infections, particularly the black fungus, *Mucormycosis*, have

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drawn global attention [4]. Endogenous fungal infection posed by *Candida* groups [6] and exogenous infection through fungi like *Cryptococcus* and *Aspergillus* had become a significant problem for people with immunity problems [7]. The black fungus, scientifically known as *Mucormycosis*, is an aggressive, severe, and rare fungal infection affecting several pre and post COVID-19 patients [8]. Among the *Candida* species, *Candida albicans* is the fourth most common nosocomial opportunistic commensal in the human body [9]. *C. albicans* mainly targets skin, gastric and urinogenital system. It is also associated with candidiasis, tuberculosis, biofilm formation, and several infections. The *C. albicans* have developed drug resistance mechanisms with a repertoire of strategies to overpower the effects of various classes of drugs [10]. The opportunistic *C. albicans* and hospital-acquired infections claim millions of lives every year worldwide. So, there is an imperative need to understand the molecular mechanism to develop effective treatment modalities [11]. Azole drug-resistant *Candida* infections reduce treatment options, and patients with candidemia are less likely to survive [12]. Another *Candida* pathogen, *Candida auris*, resistant to fluconazole, is a significant health issue globally [13].

There are different antifungal drugs. They belonged to four major classes 1. Azole class [imidazoles varieties, triazoles derivatives, and posaconazole species] 2. Echinocandins [Caspofungin, micafungin, and anidulafungin] 3. Polyenes [nystatin and amphotericin B] 4. Flucytosine. The azole drugs inhibit the ERG11 gene-derived L anosterol 14-α-demethylase enzyme involved in drug intake. It affects the biosynthesis of ergosterol and functions of drug efflux pumps Mdr1p and Cdr1p/Cdr2p. Echinocandins inhibit glycan synthesis. Polyenes bind with ergosterol and affect the efflux pump. Flucytosine inhibits the activity of DNA/RNA synthesis. Of the four classes of antifungal drugs, as the azole groups are used much, and drug resistance to azole drugs is high. In the case of azole-resistant, *C. albicans* the molecular level changes in drug resistance involve mutation of drug target genes ERG11 and ERG3and over-expression of ERG11, and up-regulation of drug efflux genes CDR1 and CDR2 [9,15,16]. In *C. albicans*, one of the factors for fluconazole resistance is overexpression of the MDR1 gene. It encodes a multidrug efflux pump [17]. It is reported that in *C. albicans*, MDR1 and MRR1 genes are associated with efflux-related resistance to fluconazole drugs [18,19]. Also, Gain-of-function mutations [GOF] in the transcription factors MRR1 and TAC 1 are responsible for overexpression of the MDR1 and CDR1/CDR2 efflux pump in fluconazole-resistant *C. albicans* [9,20]. Inactivation of MDR1 was reported to reduce the virulence of *C. albicans* [21].

The role of MDR1in drug-resistant *C. albicans* is proved. As MDR1genes encoding the regulator protein gets affected during drug resistance in *C. albicans*, in the present study, attention is focused on the sequence of the MDR1 genes to confirm any changes in the base pair of the MDR1 gene drug-resistant *C. albicans*.

Materials and methods

For the study, sputum samples of 100 patients [male in the age group 43–65 y], diagnosed with tuberculosis and taking the drugs for TB for over a year were collected from a diagnostic Centre in Tamilnadu after getting written consent from the participants. After receiving approval from the Institutional Review board ethics committee [IAEC/BIO/Re/36/1021/CPCSEA Prathyusha engineering college, India] following the Declaration of Helsinki (2017-KAEK-189_2019.02.28_18), the investigation was started. The swabs were streaked onto Sabouraud dextrose agar culture plates with chloramphenicol and incubated for 48 h at 27 °C. Isolated *Candida* colonies on SDA were subculture using *C. albicans* specific media and incubated at 37 °C for 24 h. *C. albicans* are identified based on morphological characters and by using Vitek 2 yeast identification system. The identified *C. albicans* was tested for drug resistance. Drug resistance was probed using the procedure of Clinical and Laboratory Standards Institute (CLSI-M44-A) [22].

Antifungal susceptibility testing

From a 24-h old culture of *C. albicans* in the SDA culture plate, five to seven colonies were inoculated into 5 ml of sterile saline, with turbidity adjusted to 0.5 McFarland standards. A sterile cotton wool swab soaked in the adjusted inoculum suspension was streaked on the Müller-Hinton agar (MHA) surface to prepare a lawn of the isolate. The susceptibility to the drugs was tested using the disk diffusion method. The standard antifungal discs (Thermo Scientific Oxoid) representing voriconazole (10 μg),fluconazole disk (10 μg), clotrimazole (10 μg), itraconazole (10 μg), voriconazole (10 μg), and nystatin (100 IU) were applied on Muller Hinton Agar using disc dispenser according to Clinical Laboratory Standard Institute (CLSI) M44A document.

The plates were incubated at room temperature at 35 °C for 24 h. The diameters of zones of inhibition were measured in mm. The antifungal susceptibility criteria, susceptible S, susceptible dose-dependent SDD, and resistant R was interpreted according to CLSI standards. For positive control *C. albicans* (ATCC) 90028 was used.

DNA isolation

A freshly grown culture of *C. albicans* (DCAI) showing resistance to all the five drugs tested was chosen [2 ml]. The culture was centrifuged at 10,000 rpm for10 min. After removing the supernatants, the pellets were again suspended in 465 μl of TE buffer. To the content, 30 μl 10%SDS and 3 μl of 20 mg/ml proteinase K were introduced and mixed well. After mixing, the contents were incubated for 2 h at 37 °C followed by the addition of an equal volume of phenol: chloroform mixture and mixed gently. The mixture was then centrifuged at 12,000 rpm for 10 min, and the upper aqueous phase was transferred into an Eppendorf tube. Then phenol: chloroform with equal volume was added to the Eppendorf tube and mixed uniformly. The mixture in the Eppendorf tube was further centrifuged at 12,000 rpm for 10 min. The aqueous phase was transferred to a fresh microfuge tube, and 1/10 vol of it was added and mixed until the DNA got precipitated. After precipitation, the mixture was centrifuged at 12,000 rpm for 10 min. The pellets in the tube are DNA molecules. Using 1m1 of 70% of ethanol, the pelletized DNA was washed twice for 30 s. The pellets were then dried to remove ethanol at 37 °. The ethanol was removed, and the DNA was dried at 37 °C. After suspending the dried DNA pellets in 50 μl TE buffer, DNA was stored at −20 °C till further use.

MDR1 gene sequencing

MDR I gene that encodes the drug efflux protein in membrane transport in the drug-resistant *C. albicans* isolates was chosen. The primers were designed based on that target. The designed primer is given below.

MDR1 Forward — 5′ AGGTGAACCCAATTCAAGTC 3′

MDRI Reverse — 5′ ACAACTGGTTCCATAACGGT 3′

The DNA from MDRI isolates was separated and subjected to a PCR study [VeritiPro Thermal Cycler-Thermo Fisher] to sequence the gene. In a 25 μl PCR vial, the template DNA, primer for MDR1 (1 μl), and other reagents were added and mixed well. The PCR vials were placed into the thermal cycler. After the first cycle of denaturation at 95 °C., for 30 s, 30 cycles of annealing at 57 ° C for I min, and elongation at 72 ° C for I min was carried out. As reported, the amplified DNA was visualized using agarose gel electrophoresis with a marker DNA [Fermentas, USA] [23]. For sequencing, Sanger

Table 1
Susceptibility profiles of *Candida albicans* against different antifungals [Mean ± SD N/3]. The drug sensitivity prevalence percentage in the clinical isolates is given in parenthesis.

No.	Name of the antifungals	Diameter of zone of inhibition (mm)						
		S		SDD		R		Control ATCC strain-90028
		CLS	A	CLS	A	CLS	A	
l.	Nystatin100U	≥**15**	17.5 ± 0.3 (**21.17**)	10–14	11.5 ± 0.4 (37.65)	≥**10**	6.20 ± 0.6 (41.18)	24.0
2.	Clotrimazole 10 μg	≥**20**	21.45 ± 0.4 (8.24)	12–19	13.8 ± 0.5 (29.41)	≥11	7.9 ± 0.6* (62.35)	26.5
3.	Fluconazole 10 μg	≥**19**	20.50 ± 0.5 (5.88)	15–18	16.8 ± 0.3 (23.53)	≥14	9.5 ± 0.6* (70.59)	30.5
4.	Itraconazole 10 μg	≥**19**	20.75 ± 0.50 (12.94)	15–18	16.40 ± 0.5 (28.24)	≥14	10.5 ± 0.6* (58.82)	29.7
5.	Ketoconazole 10 μg	≥**15**	17.60 ± 1.20 (15.29)	10–14	11.40 ± 0.7 (37.65)	≥9	7.1 ± 0.6* (47.06)	30.4

CLSI interpreted antifungal drugs — susceptible [S], susceptible dose-dependent [SDD], and resistant (R) values according to CLSI standards. A — actual value obtained in the present study.

* Significantly resisted — $p < 0.05$.

Sequencing 3500 Series Genetic Analyzers [Thermo Fisher] were used.

Results

In the present study, one hundred *C. albicans* isolates collected from chronic tuberculosis patients were clinically tested. Among them, 85% that showed resistance to three or more antifungal drugs are categorized as multidrug-resistant [MDR] fungal species [Table 1]. Of the five antifungal drugs tested, four belonged to the azole family, and nystatin belonged to the polyene class. Except for nystatin, all the other azole drugs are highly resisted by the *C. albicans* tested. For all the four azole drugs, the diameter of the zone of inhibition exhibited by *C. albicans* isolates is more or less near, indicating the development of resistance in the entire class of azole antifungals. The diameter of zone of inhibition for sensitive [S], susceptibility dose-dependent [SDD], and resistant [R] was recorded as per CLS protocol for the individual drug.

Susceptibility of *C. albicans* concerning nystatin shows that nystatin was sensitive to 21. 17% [18/85] of the isolates, 37.65% [32/85] of the isolates were with SDD, and the remaining 41.18% [35/85] of the isolates were resistant to nystatin. The mean diameter of the zone of inhibition seen for nystatin sensitive, SDD, and Resistant cases were 17.5 ± 0.3, 11.5 ± 0.4, and 6.20 ± 0.6 mm, respectively. The Clotrimazole drug was sensitive to 8.24% [7/85] cases with a mean inhibition zone of 21.45 ± 0.4 mm, 29.41% [25/85] for susceptible dose-dependent [SDD] isolates with a mean zone diameter of 13.8 ± 0.5 mm. For drug-resistant isolates, the mean zone diameter was 7.9 ± 0.6 mm in 62.35% [53/85] cases. The susceptibility pattern of *C. albicans* to fluconazole shows that only5.88% [5/85]of them were sensitive with an inhibition zone of 20.50 ± 0.5 mm. Further, 23.53% [20/85] of the isolates were dose-dependent [SDD] with a mean inhibition zone diameter, 16.8 ± 0.3 mm. In 70.59% of the fluconazole-resistant cases, the mean diameter of the inhibition was 9.5 ± 0.6 mm. The Itraconazole susceptible isolates 12.94% [11/85] were sensitive with a mean diameter of 20.75 ± 0.50 mm inhibition zone. In 28.24% [25/85] susceptible dose-dependent [SDD] isolates, the mean inhibition zone diameter was 16.40 ± 0.5 mm, and the diameter was 58.82% of the drug-resistant isolates of the inhibition zone was 10.5 ± 0.6 mm. The mean diameter of the zone of inhibition was 17.60 ± 1.20 mm in 15.29% [13/85 isolates] of Ketoconazole-sensitive isolates. In 28.24% [32/85] isolates with SDD nature, the mean diameter of the inhibition was 11.40 ± 0.7 mm. The mean diameter of the zone of inhibition was 7.1 ± 06 mm in 47.06% [40/85] Ketoconazole resistant isolates.

For the MDRI gene sequencing, the isolates showing high resistance to all the five drugs were chosen. As the tested fungal strain, was resistant to all antifungal agents. The resistant genes in the strain were chosen as MDR strain for further evaluation. From the chosen *C. albicans* (DCAI), the MDR1 gene was chosen for amplification. The obtained gene product has 835 bp [Fig. 1]. The observed *sequence* pattern of MDR1 promoters from the isolates was compared with three similar sequences deposited in Gen Bank using BLAST. Their accession number is DQ903899, DQ903900, and Y1470. Using Clustral W2 multiple sequences alignment was made. The base pair arrangements from 132 to 835bp in the MDR1 gene of DCAI were compared with that of the Gen bank data. Variations were noted at positions 466, 6l9, 820, 82l, 822, 833, and 834 of the base pairs. These variations in the base pair sequence indicate mutational changes in the MDR1 gene leading to drug resistance *C. albicans* [Fig.2].

Discussion

Fungal strains are potent challengers to human health like bacteria. Every year more than 120 million cases of severe fungal infections are reported worldwide, and 1.7 million deaths per year are reported to be fungal invasions [24]. The victims of fungal infections are mostly immune weakened or immunocompromised due to various health issues, including COVID infection. In Saudi Arabia, the threat from invasive fungal infections in hospitalized, immunocompromised and other cases are reported [25–27]. In the present study, the fungal strain *C. albicans* isolated from tuberculosis-infected males' sputum was analyzed for drug resistance assay. The sensitivity of the isolate to five antifungal drugs, Nystatin, Clotrimazole, Fluconazole, Itraconazole, and Ketoconazole, showed significant variations over the control. The positive control ATCC 90028 *C. albicans* strain was sensitive to all the drugs tested. This observation shows that the clinical isolates have developed drug resistance. The clinical isolates with drug resistance were due to their inhabitation in patients with chronic tuberculosis medication, as reported [28]. Among the 100 clinically positive samples, eighty-five swabs showed resistance to more than three antifungal agents [85.00%].In the 85 positive cases, a differential pattern of resistance to the drugs was seen. The resistance to the drug nystatin in the sampled population is considerably less. Among the five tested drugs, the isolates with sensitivity for nystatin were high [21.17%]. Among the different azole-resistant isolates, sensitivity was high [15.29%] for Ketoconazole. The drug resistance was maximum for fluconazole [70.59%]. The results indicate that more than 75% of the clinical isolates are in the intermediate [SSD] and resistant[R] range. For all the four azole drugs, the diameter of the zone of inhibition exhibited is more or less closely, confirming the development of resistance to the entire class of azole antifungals. The development of resistance to azole class drugs may be due to several molecular adaptations to withstand the persistent inhibition by these classes of drugs over time. The reduction in sensitivity of the *C. albicans* isolates to the azole drugs indicates that *C. albicans* is progressing towards total azole-resistant pathogens. So urgent

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Y14703
TTTGCTAATCCTTGTTTGGCTACTGGTGGTGCAAGTGTTGCTGATGTGGTTAAATTTTGG720
DQ903889
TTTGCTAGTCCTTGTTTGGCCACTGGTGGTGCAAGTGTTGCTGATGTGGTTAAATTTTGG717
DQ903900
TTTGCTAGTCCTTGTTTGGCCACTGGTGGTGCAAGTGTTGCTGATGTGGTTAAATTTTGG717
DCAL1
TTTGCTAGTCCTTGTTTGGCCACTGGTGGTGCTAGTGTTGCTGATGTGGTTAAATTTTGG493[466]

Y14703
TGTATCATTTCTGCGTTTTCATTTGTTATGTTGTGTTTCACTTTACCTGAAACTTTTGGC 900
DQ903889
TGTATCATTTCTGGGTTTTCATTTGTTATGTTGTGTTTCACTTTACCTGAAACTTTTGGC 897
DQ903900
TGTATCATTTCTGGGTTTTCATTTGTTATGTTGTGTTTCACTTTACCTGAAACTTTTGGC 897
DCAL1
TGTATTATTTCTGGGTTTTCATTTGTTATGTTGTGTTTCACTTTACCTGAAACTTTTGGC 673[619]

Y14703
TTATGGAGGCCATTAGAAATCACCGTTATGGAACCAGTTGTTCTTTTGATTAACATTTAC1080
DQ903889
TTATGGAGACCATTAGAAATCACCGTTATGGAACCAGTTGTTCTTTTGATTAACATTTAC1077
DQ903900
TTATGGAGACCATTAGAAATCACCGTTATGGAACCAGTTGTTCI1IIGATTAACATTTAC1077
DCAL1
TTATGGAGACCATTAGAAATCACCGTAGGGGAAACCGTTTGT835[820,821,822,833,834].
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Fig. 1. The sequence of MDRl of CAD1 compared with three similar sequences compared from Gen Bank using BLASTN. Seven variations at the positions of bp466, 6l9, 820, 82l, 822,833, and 834 indicate mutation in the MDR1 gene of *C. albicans* isolated.

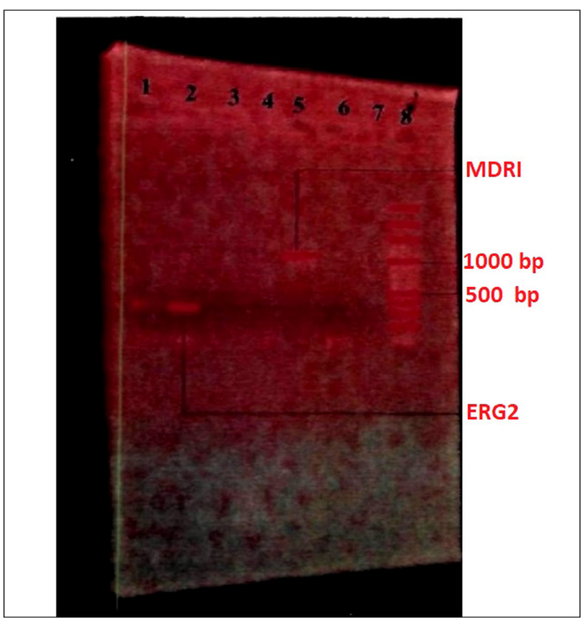

Fig. 2. An image of 1% agarose gel indicating amplified gene fragments. A molecular references.

attention is needed in this era of mounting fungal infections in the lungs.

Antimicrobial drugs are generally designed to intercept metabolic pathways like nucleotide synthesis and inhibit DNA/RNA and protein synthesis. Antifungal drugs also disrupt the cell membrane and enzymes like chitin synthase involved in cell wall synthesis. The azole class drugs execute antifungal action by inhibiting the lanosterol–ergosterol synthesis and making changes in the fluidity and agility of the membrane bounding the fungal strains [29].

When the *C. albicans* get transformed into a drug-resistant form, a mutation in the ERG gene occurs. As a result, the binding of azole drugs with targeted fungi fails. Further, the overexpression of the efflux pump pushes out the drug molecules. Also, the mutation occurs in the ERG3 gene, leading to the upregulation of the enzyme target and the bye-pass pathway [30].

The emergence of drug resistance favours invasive *Candidiasis* infections, leading to morbidity and mortality and leads to a global economic burden. With the advent of new resistance mechanisms, the efficiency of treating common infectious diseases may fail. The loss of microbial response to standard treatment will prolong the illness, increase hospital treatment expenditures, and give risk to the patient's life. The abuse, misuse, and overuse of azole drugs for many common skin fungal ailments favour the *C. albicans* to immunize the drug's effect. A continuous drug pressure and the struggle for existence induce the threatened fungal species to develop molecular adaptation to resist the drugs, leading to the further spread of MDR [15]. The invasion of drug-resistant *C. albicans* promotes biofilm formation in catheters and other implants with drug-resistant microbial colonization. The drug-resistant *C. albicans* pose a serious threat to immunocompromised cases, chemotherapy receivers, HIV patients, invasive candidiasis, SARsCo -19 victims, and also pulmonary mycosis patients [31]. A greater global health issue may arise if pulmonary mycoses due to *C. albicans* and another opportunistic fungal pathogen with drug resistance is neglected [29]. The epidemiology of the drug-resistant *C. albicans* has become a matter of great concern. The modifications in the molecular functions in *C. albicans* concerning the development of azole drug resistance need to be addressed. In azole-resistant *C. albicans*, it is reported that the upregulates the genes *CDR1* and *CDR2* related to efflux pump. Combining azole drugs with oridonin remediated the drug resistance in candidiasis infection with oridonin (ORI) [32].

There is different adaptive molecular evolution in *C. albicans* to overcome their frequent killer belonging to the azole family of antifungal drugs [28]. The gene overexpression and an increase in the activity of proteins that reduce oxidative damage are some of the reasons for drug resistance [33,34]. The increased expression of ERG11gene interfered with the efflux mechanism in the cell boundary leading to the expulsion of the drugs and inhibiting the drug's action. Mutations in the *ERG11* gene result in reduced affinity of *ERG11* to fluconazole leading to the inhibition of its power.

In *Candida* species, two gene transporter families, the CDR genes and MDR genes are involved in the development of drug resistance. The Cdr1 and Cdr2 genes of the CDR family use energy derived from the hydrolysis of ATP to transport drugs outside the cells [35]. But MDR1 belonged to the major facilitator superfamily genes gets overexpressed in the drug-resistant process. It interferes with membrane transport protein synthesis [36], and fluconazole is pumped out of the cell to reduce intracellular accumulation and antibiotic stress [37]. MDR1 utilizes energy from the proton gradient for drug extrusion [38]. The MDR1 gene overexpression and mutations in ERG11 are responsible for high-level fluconazole resistance [39].

The inactivation of overexpressing MDR1 in *C. albicans* increased the susceptibility of *C. albicans* to fluconazole [35].In the present study, the sequencing of MDR1genes isolated from azole drug-resistant *C. albicans* showed variations in the nucleotide base pairs at seven sites indicating changes in MDR1genes leading to efflux-mediated antifungal drug resistance. Experimental studies show that deleting *the MDR1* gene from *MDR1*-overexpression in *C. albicans* isolates leads to a decrease in fluconazole resistance [38].

Conclusion

In the present study, the fungal strain *C. albicans* isolated from tuberculosis-infected males' sputum was analyzed for drug resistance assay. In *C. albicans*, resistance to the drugs nystatin, clotrimazole, fluconazole, itraconazole, and Ketoconazole was observed. A comparison with the positive control ATCC 90028, *C. albicans* strain showed that only the clinical isolates had lost their sensitivity to all the drugs tested. The clinical isolates have developed resistance mechanisms because of their inhabitation in patients with chronic tuberculosis medications as *C. albicans* co-exist with tuberculosis pathogens. The current study informs the widespread drug resistance in *C. albicans* as an evolutionary adaptation to survive in their struggle for existence during azole, antifungal stress. So, it is imperative to treat fungal infections with alternative drugs or drugs admixed with azole derivatives.

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Competing interests

None declared.

Ethical approval

Required and Provided.

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